
Session 5 (2h) — Capstone: Sensor Enclosure + Export

Goal

Combine everything from Sessions 1–4 into one small, realistic deliverable: a two-part enclosure (base + lid) sized to hold an Arduino Uno/Nano and one sensor, ready to hand off to the `3d-printing-prep/` pack.

What it must include

- Fully constrained sketches (Session 1)
- Extrude + shell + fillets for the enclosure body (Session 2)
- Base + lid as separate components joined together (Session 3)
- At least 3 parameters driving key dimensions, with a defined tolerance on the mounting holes (Session 4)

Build steps (1.5h)

1. Measure (or look up) your Arduino board's outline + mounting-hole positions.
2. Sketch the base footprint with parametric `wall_thickness`, `standoff_height`.
3. Extrude + shell the base; add 4 internal standoffs (small cylinders) at the board's mounting holes, sized with your tolerance rule from Session 4.
4. Add a cutout on one wall sized for a USB cable / sensor wire pass-through.
5. Model the lid as a separate component; join it to the base (Rigid or Slider — your choice) and check interference.
6. Fillet all external edges (2mm) for print quality.

Export (0.5h)

1. Right-click each component (base, lid) → **Save As Mesh** → export as **STL** (or 3MF).
2. Use a reasonable mesh refinement (Fusion's "High" preset is enough for FDM printing — don't export at max resolution, it just bloats file size for no visible benefit).
3. Save both STL files somewhere findable — these become the direct input to `3d-printing-prep/session-01`.

Checkpoint

- ☐ Enclosure has a base + lid as separate, joined components
- ☐ Uses parameters for at least wall thickness and a mounting-hole tolerance
- ☐ Both parts exported as STL, ready for slicing
- ☐ Can point to where sketching / extrude+shell / assembly-joint / parametric each happen in the finished file